Practice Problems

1. In humans, a widow's peak hairline is dominant (W) over a straight hairline (w). If two heterozygous parents both have widow's peaks, what is the probability their child will have a straight hairline? Could two straight-haired parents ever have a child with a widow's peak? Why or why not?
2. In a breed of cattle, the presence of horns is determined by a single gene. The allele for being polled (naturally hornless, P) is dominant to the allele for having horns (p). A rancher crosses a true-breeding polled bull with a true-breeding horned cow.
   a) What percentage of the F1 calves will have horns?
   b) If the rancher then crosses two of these F1 cattle with each other, what is the expected phenotypic ratio (polled : horned) among the F2 generation calves?
3. A certain breed of dog can have either floppy ears or pointed ears. When two floppy-eared dogs are crossed, their litters always contain both floppy-eared and pointed-eared puppies in a ratio of approximately 3 floppy : 1 pointed. A cross between a floppy-eared dog and a pointed-eared dog produces equal numbers of floppy and pointed puppies. What is the most probable mode of inheritance for ear type in these dogs? Draw a Punnett Square crossing two floppy-eared dogs to explain the observed 3:1 ratio.
4. In cats, short hair (L) is dominant over long hair (l). A long-haired cat is bred with a heterozygous short-haired cat. What are the genotypic and phenotypic ratios of their offspring?
5. You are a pea plant breeder. You have a plant with purple flowers (dominant trait, P) but don't know if it's homozygous (PP) or heterozygous (Pp). Describe a test cross you would perform to determine its genotype and predict the possible outcomes.
6. In the dog breed Labrador retrievers, black coat (B) is dominant to chocolate (b). At a different independently assorting locus normal vision (N) is dominant to progressive retinal atrophy (PRA) (n). A dog breeder crosses a dog heterozygous for both traits (BbNn) with a chocolate Lab that carries PRA (bbNn). What is the probability of getting a chocolate puppy with normal vision?
7. A geneticist crosses two pea plants with purple flowers. To her surprise, the offspring include both purple-flowered and white-flowered plants in a ratio of 3 purple : 1 white. She then crosses a purple-flowered offspring plant with a white-flowered plant. This cross yields offspring in a 1 purple : 1 white ratio. What do these results tell you about the genotypes of the original purple-flowered parent plants? Propose a single-gene model to explain both results.
8. A gardener crosses tomato plants: one is true-breeding for tall vines and red fruit (TTRR), the other for dwarf vines and yellow fruit (ttrr). The F1 plants are all tall with red fruit. If the F1 plants are intercrossed, what fraction of the F2 plants would you expect to be tall with yellow fruit? Explain how this demonstrates independent assortment.
9. Cystic fibrosis (CF) is an autosomal recessive disorder (f allele). The ABO blood group is on a different chromosome. A couple are both carriers for CF (Ff) and the man has blood type A (I^Ai) while the woman has type B (I^Bi). What is the probability their child will have CF and blood type O?
10. A farmer breeds chickens and observes two traits: comb shape (single vs. rose) and leg feathering (clean vs. feathered). He crosses a single-combed, clean-legged bird with a rose-combed, feathered-legged bird. All F1 offspring are rose-combed with

feathered legs. Crossing two F1 birds produces an F2 generation with four phenotypes in the following proportions: 9 rose/feathered : 3 rose/clean : 3 single/feathered : 1 single/clean. What fundamental Mendelian principle do these results demonstrate? How many genes are involved, and what are the dominance relationships for each trait?

11. Labrador coat colour is governed by two genes: the B gene (B=black, b=chocolate) and the E gene. The E gene is epistatic: if a dog is homozygous recessive (ee), it is yellow regardless of its B gene genotype. Two black Labs, both heterozygous for both genes (BbEe), are crossed. What are the expected phenotypic ratios among their puppies (black:chocolate:yellow)?
12. In squash (a type of vegetable), fruit colour is controlled by two genes. Gene W allows colour (W-) or prevents it (ww, resulting in white). Gene G determines green (G-) or yellow (gg) only if colour is allowed. A white squash (wwGG) is crossed with a yellow squash (WWgg). What is the phenotype of the F1? If two F1 plants are crossed, what is the probability of getting a white squash in the F2?
13. In mice, agouti (banded) fur (A) is dominant to solid colour (a). A second gene, C, allows pigment to be deposited; cc mice are albino regardless of their A gene genotype. A mouse heterozygous for both genes (AaCc) is crossed with a solid-coloured, albino mouse (aacc). What are the expected phenotypes and their ratios in the offspring?
14. In a new flower species, a botanist crosses two true-breeding white-flowered plants. The entire F1 generation is purple. Self-pollinating these F1 plants yields an F2 generation with a phenotypic ratio of 9 purple : 7 white. What specific type of gene interaction best explains these results? Propose a two-gene model (using letters A/a and B/b) and state the genotypes of the original white parents and the purple F1 plants.
15. In sweet peas, purple flower colour requires two dominant alleles: C (for colour) and P (for purple pigment). Plants with genotype cc or pp are white. Two white-flowered plants, one with genotype CCpp and the other with genotype ccPP, are crossed. What colour are the F1 flowers? What is the phenotypic ratio in the F2 generation after an F1 self-cross?
16. In a hypothetical chicken breed, two traits follow non-Mendelian patterns at separate genes. Feather color shows incomplete dominance at the Color (C) locus: Black ($C^B C^B$) and White ($C^W C^W$) parents produce Blue ($C^B C^W$) offspring. Comb shape shows codominance at the Shape (S) locus: Pea comb ($S^P S^P$) and Rose comb ($S^R S^R$) parents produce Walnut comb ($S^P S^R$) offspring. A blue-feathered, walnut-combed rooster ($C^B C^W$, $S^P S^R$) is crossed with a black-feathered, pea-combed hen ($C^B C^B$, $S^P S^P$). Describe the genotypic and phenotypic ratios of their chicks.
17. Assume hair texture shows incomplete dominance: Curly (C1C1), Wavy (C1C2), Straight (C2C2). The M-N blood group is codominant: M ($L^M L^M$), MN ($L^M L^N$), N ($L^N L^N$). A man with wavy hair and M-type blood (C1C2, $L^M L^M$) marries a woman with straight hair and MN-type blood (C2C2, $L^M L^N$). List all possible phenotypes of their children and the probability of each.
18. In cattle, a breeder notices that crossing a "roan" cow (a mixture of red and white hairs) with a solid red bull produces calves that are either roan or red in equal numbers. Crossing two roan cattle produces calves that are red, roan, and white in a 1:2:1 ratio. What is the genetic relationship between the alleles for red, white, and

roan coat colour? Name the specific inheritance pattern and explain the 1:2:1 ratio in the second cross.

19. In snapdragons, red (RR) and white (rr) flowers produce pink (Rr) offspring. Leaf shape shows incomplete dominance: broad (BB), narrow (bb), and intermediate (Bb). A pink-flowered, intermediate-leaved plant is crossed with a white-flowered, broad-leaved plant. What fraction of the offspring will have pink flowers and intermediate leaves?
20. A plant breeder is working with a species where flower size shows incomplete dominance (Large=LL, Medium=Ll, Small=ll) and petal edge shows codominance: smooth (AA), serrated (aa), and both smooth & serrated (Aa). If a medium, smooth/serrated plant (Ll, Aa) is self-pollinated, what is the chance of getting a small-flowered plant with serrated edges?
21. The Manx tailless allele (M) in cats is dominant. However, homozygous embryos (MM) die before birth. Two Manx cats (both Mm) are crossed. What is the expected phenotypic ratio (tailless : tailed) among their surviving kittens? Why does this deviate from a normal monohybrid ratio?
22. A mouse breeder crosses two mice with normal-length tails. Their litter results are puzzling: about 2/3 of the pups have normal tails, and 1/3 have very short, kinked tails. When the breeder crosses a normal-tailed mouse from such a litter with a short-tailed mouse, the offspring show a 1 normal : 1 short-tailed ratio. The 2:1 ratio among offspring of normal-tailed parents is a classic indicator of a specific genetic phenomenon. What is it? Propose a single-gene model with genotypes to explain all observations. Why don't you see the expected Mendelian 3:1 ratio?
23. An allele for yellow coat colour (A^Y) in mice is dominant to agouti (A). Mice homozygous for the yellow allele ($A^Y A^Y$) die in utero. What are the expected genotypic and phenotypic ratios from a cross between two yellow mice ($A^Y A$)?
24. In a certain ornamental plant, a dominant allele (F) causes fringed leaves. The homozygous dominant (FF) is lethal at the seedling stage. A botanist crosses a fringed-leaf plant with a normal plant. Half of the resulting seedlings die. What was the genotype of the fringed-leaf parent? Explain.
25. In rabbits, two pure-breeding lines exist: one with long, black fur and another with short, white fur. Crossing these produces F1 rabbits all with long, black fur. Intercrossing the F1 yields an F2 generation with a surprising distribution: 9 long/black : 3 long/brown : 3 short/black : 1 short/brown. The appearance of two new colours (brown fur and a new pattern of short/brown) suggests interaction between two genes. What is the most probable mode of inheritance for fur length and colour? Explain the appearance of the "brown" phenotype in the F2.
26. Paternity Puzzle: A woman with blood type O has a child with blood type O. She claims a man with blood type AB is the father. Is this genetically possible? Explain using allele combinations.
27. In a population of fruit flies, researchers are studying two traits: wing shape (normal vs. curled) and eye color (red vs. purple). A cross is performed between a fly that is true-breeding for normal wings and purple eyes, and a fly that is true-breeding for curled wings and red eyes. All F1 offspring have normal wings and red eyes. When these F1 flies are intercrossed, the F2 generation produces the following phenotypic ratio among hundreds of offspring:
    Normal wings, Red eyes: 142
    Normal wings, Purple eyes: 41

Curled wings, Red eyes: 45
Curled wings, Purple eyes: 12
What does this observed ratio indicate about the inheritance of these two traits?

28. In a family, the mother has blood type B and the father has blood type A. They have four children: one with type A, one with type B, one with type AB, and one with type O. What must the genotypes of the parents be? Show your work.
29. Forensic analysts are trying to identify the source of a bloodstain (Sample X) found at a crime scene. Sample X tests positive for both A and B antigens. Suspect 1 tests as blood type AB. Suspect 2 tests as blood type A, but genetic testing reveals she carries one I^A allele and one i allele. Could Sample X belong to Suspect 2? Why or why not, based on the genetics of the ABO system?
30. Read the explanation below and then answer the question.
    The H locus (FUT1 gene) and the ABO locus are on different chromosomes (19 and 9), but they are functionally linked in a biochemical pathway for blood antigen synthesis:
    a. The H gene produces an enzyme that creates the H antigen (a carbohydrate precursor) on red blood cells.
    b. The ABO gene produces enzymes that modify the H antigen:
        i. The I^A allele adds N-acetylgalactosamine → A antigen.
        ii. The I^B allele adds galactose → B antigen.
        iii. The i allele adds nothing → H antigen remains unmodified (type O phenotype).

    This creates a hierarchical epistasis:
    If the H antigen is absent (genotype hh at the FUT1 locus → Bombay phenotype), the ABO enzymes have no substrate to act on, so no A or B antigens are produced regardless of the ABO genotype. The person will test as type O serologically. However, the Bombay blood group person (hh) can produce potent anti-H antibodies against the H antigen.
    Question:
    A couple is puzzled by their child's blood type. The mother has type A blood. The father has type B blood. Their child has type O blood. All three individuals undergo genetic testing. The child is found to have the I^A allele from the mother and the I^B allele from the father, yet serologically tests as type O. What genetic phenomenon can explain a child who inherits I^A and I^B alleles but has type O blood? Name the phenomenon and describe the required genotype at a second locus to produce this result.